Discrete - 1.1.63

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Question

Quadrature formula:
$\int_0^1 f(x)\,dx \approx \frac{1}{12}[f(0) + 2bf(0.25) + 2f(0.5) + 2df(0.75) + f(1)]$.
If used as Simpson's $\frac{1}{3}$ rule: (2007 XE)

a) b = d = 1 b) b = d = 2 c) b = 2d = 1 d) b = 2d = 2

Simpson's rule

Simpson's 1/3 rule is a numerical integration method that approximates the area under a curve by fitting parabolic arcs (second-order polynomials) through sets of three points.
For a single interval $[x_0, x_2]$ with a midpoint x_1 and step size h:

$$\int_{x_0}^{x_2} f(x)\,dx \approx \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2)\right] \tag{1}$$

In general Simpson's 1/3 Rule over n intervals (where n must be even):

$$\int_a^b f(x)\,dx \approx \frac{h}{3}\left[f(x_0) + 4\sum_{i=1,3,5}^{n-1} f(x_i) + 2\sum_{j=2,4,6}^{n-2} f(x_j) + f(x_n)\right] \tag{2}$$

Where $h = \frac{b-a}{n}$ is the uniform step size and $x_0 = \text{a}$, $x_1 = \text{b}$.

Solution

The problem provides a specific quadrature formula:

$$\int_0^1 f(x)\,dx \approx \frac{1}{12}\left[f(0) + 2bf(0.25) + 2f(0.5) + 2df(0.75) + f(1)\right] \quad (3)$$

Based on the function arguments $\{0, 0.25, 0.5, 0.75, 1\}$:

- $a = 0, b = 1$
- $n = 4$ intervals
- $h = \frac{1-0}{4} = 0.25$

Substituting $h = 0.25$ into the general Simpson's Rule for $n = 4$:

$$\int_0^1 f(x)\,dx \approx \frac{0.25}{3}\left[f(0) + 4f(0.25) + 2f(0.5) + 4f(0.75) + f(1)\right] \quad (4)$$

Simplifying $\frac{0.25}{3} = \frac{1}{12}$, the standard internal weights are:

$$\text{Internal Weights} = [4, 2, 4] \quad (5)$$

Solution

Comparing these to the coefficients inside the brackets of the equation (3) we get

- Coefficient for $f(0.25)$: $2b = 4 \implies \mathbf{b = 2}$
- Coefficient for $f(0.75)$: $2d = 4 \implies \mathbf{d = 2}$

Numerical Example

Lets test the above relation with $f(x) = x$

Exact Integral :

$$\int_0^1 x\,dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2} = 0.5 \tag{6}$$

Applying the Formula

$$\int_0^1 f(x)\,dx \approx \frac{1}{12}\left[f(0) + 4f(0.25) + 2f(0.5) + 4f(0.75) + f(1)\right] \tag{7}$$

Solution

- $f(0) = 0$
- $f(0.25) = 0.25 = \frac{1}{4}$
- $f(0.5) = 0.5 = \frac{1}{2}$
- $f(0.75) = 0.75 = \frac{3}{4}$
- $f(1) = 1$

Substitute these into above equation and simplifying it will give :

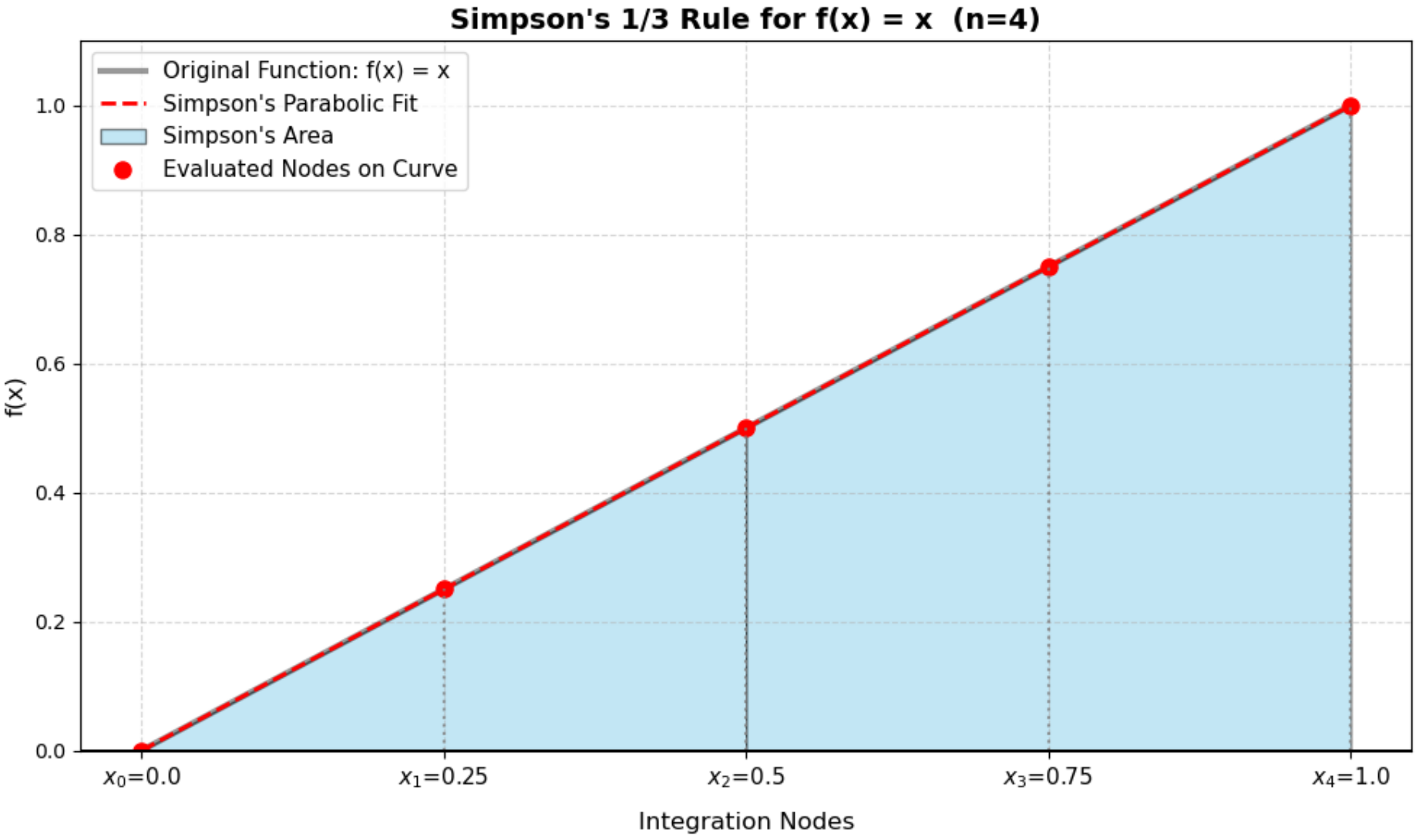
$$\frac{1}{12}\left[0 + 4\left(\frac{1}{4}\right) + 2\left(\frac{1}{2}\right) + 4\left(\frac{3}{4}\right) + 1\right] \tag{8}$$

$$\implies \frac{1}{12}[1 + 1 + 3 + 1] \tag{9}$$

$$\implies \frac{6}{12} = \frac{1}{2} \tag{10}$$

Plot

Python plot



Python: plot.py

```
import numpy as np
import matplotlib.pyplot as plt

def f(x):
    return x

a = 0.0
b = 1.0
n = 4

# Generate the nodes
x_nodes = np.linspace(a, b, n + 1)
y_nodes = f(x_nodes)

x_smooth = np.linspace(a, b, 200)
y_smooth = f(x_smooth)

plt.figure(figsize=(10, 6))
```

Python: plot.py

```
plt.plot(x_smooth, y_smooth, 'k-', label='Original
    Function: f(x) = x', linewidth=3, alpha=0.4)

#  Implement Simpson's 1/3 Rule
for i in range(0, n, 2):
    # Get the 3 points (x0, x1, x2) for this parabolic
    block
    x_triplet = x_nodes[i:i+3]
    y_triplet = y_nodes[i:i+3]

    # Fit a 2nd-degree polynomial (a parabola) to
    these 3 points
    parabola_coeffs = np.polyfit(x_triplet, y_triplet,
    2)
    parabola_func = np.poly1d(parabola_coeffs)
```

Python: plot.py

```
x_segment = np.linspace(x_triplet[0], x_triplet
[2], 50)
y_segment = parabola_func(x_segment)

# Plot the Simpson's Parabola
plt.plot(x_segment, y_segment, 'r--', linewidth=2,
         label="Simpson's Parabolic Fit" if i == 0
else "")

# Shade the area under this specific parabola
plt.fill_between(x_segment, 0, y_segment, color='
skyblue', alpha=0.5,
                 edgecolor='black', linewidth=1,
                 label="Simpson's Area" if i == 0
else "")
```

Python: plot.py

```
# Red dots on the function curve
plt.plot(x_nodes, y_nodes, 'ro', markersize=8, label='
    Evaluated Nodes on Curve')

# Add vertical dashed lines from the x-axis to the
    function
for x_val in x_nodes:
    plt.vlines(x_val, ymin=0, ymax=f(x_val), color='
    gray', linestyle=':', alpha=0.8)

# 6. Formatting the plot
plt.title("Simpson's 1/3 Rule for f(x) = x  (n=4)",
    fontsize=14, fontweight='bold')
plt.xlabel('Integration Nodes', labelpad=10, fontsize
    =12)
plt.ylabel('f(x)', fontsize=12)
plt.axhline(0, color='black', linewidth=1.5)
```

Python: plot.py

```
x_labels = [f'$x_{i}$={val}' for i, val in enumerate(
    x_nodes)]
plt.xticks(x_nodes, x_labels, fontsize=11)

plt.xlim(-0.05, 1.05)
plt.ylim(0, 1.1)

plt.legend(loc='upper left', fontsize=11)
plt.grid(True, linestyle='--', alpha=0.5)

# Show the plot
plt.tight_layout()
plt.savefig("simpson.png")
plt.show()
```